

Git Step-by-Step Commands

1. Check Git installation

```
git --version
```

Checks whether Git is installed and shows the installed version.

2. Configure username

```
git config --global user.name "Your Name"
```

Sets the name that appears in your Git commits.

3. Configure email

```
git config --global user.email "you@example.com"
```

Sets the email associated with your Git commits.

4. Create a Git repository

```
git init
```

Creates a new Git repository in the current folder.

5. Check repository status

```
git status
```

Shows changed, new, and staged files.

6. Add a file

```
git add filename.txt
```

Moves a specific file to the staging area.

7. Add all files

```
git add .
```

Stages all changes in the current directory.

8. Create a commit

```
git commit -m "Initial commit"
```

Saves the staged changes with a message.

9. View commit history

```
git log
```

Displays previous commits.

10. Clone a repository

```
git clone https://github.com/user/project.git
```

Downloads an existing remote repository to your computer.

11. Check remote repository

```
git remote -v
```

Shows the remote repository URLs.

12. Add a remote repository

```
git remote add origin https://github.com/user/project.git
```

Connects your local repository to a remote repository.

13. Push code

```
git push origin main
```

Uploads your local commits to the remote main branch.

14. Pull latest code

```
git pull origin main
```

Downloads and integrates the latest changes from the remote main branch.

15. Create a branch

```
git branch feature-login
```

Creates a new branch.

16. Switch to a branch

```
git switch feature-login
```

Moves you to the specified branch.

17. Create and switch to a branch

```
git switch -c feature-login
```

Creates a new branch and switches to it immediately.

18. List branches

```
git branch
```

Shows your local branches.

19. Merge a branch

```
git switch main
```

```
git merge feature-login
```

Switches to main and merges feature-login into it.

20. Delete a branch

```
git branch -d feature-login
```

Deletes a branch after it has been merged.

21. See unstaged changes

```
git diff
```

Shows changes that have not been staged.

22. Unstage a file

```
git restore --staged filename.txt
```

Removes the file from the staging area without deleting your changes.

23. Discard file changes

```
git restore filename.txt
```

Restores the file to its last committed state. **Be careful: uncommitted changes can be lost.**

24. Show commit details

```
git show
```

Displays information about a particular commit.

25. Save work temporarily

```
git stash
```

Temporarily stores uncommitted changes.

26. Restore stashed work

```
git stash pop
```

Brings the previously stashed changes back.

Common Git Workflow

```
git init
```

```
git status
```

```
git add .
```

```
git commit -m "Initial commit"
```

```
git remote add origin https://github.com/user/project.git
```

```
git push -u origin main
```

For future changes:

```
git status
```

```
git add .
```

```
git commit -m "Updated project"
```

```
git push
```

Git workflow to remember

Working Directory → Staging Area → Commit → Remote Repository

Create/Edit files

↓

```
git add .
```

↓

```
git commit
```

↓

```
git push
```

↓

GitHub